Lec29

13 March 2026 09:11 AM

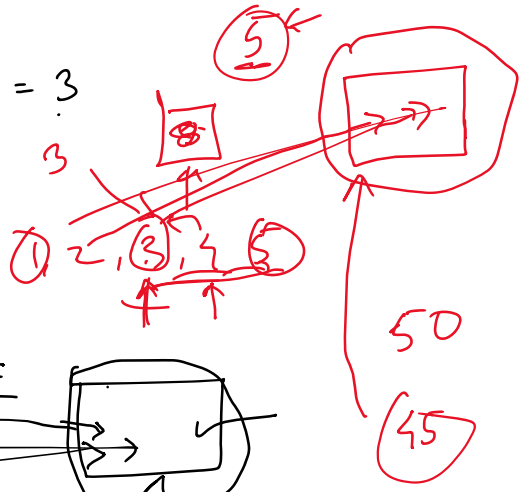
first bad version

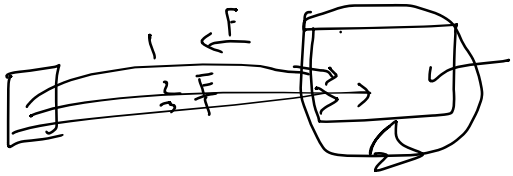
$n = 5$

bad = 3

[1, 2, 3, 4, 5]

F F T T T

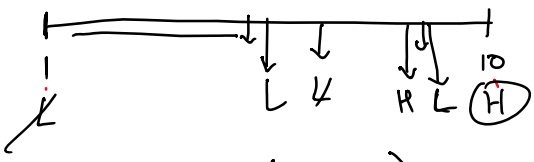




$n = \overset{5}{\cancel{100}} \longrightarrow 95$

bad = 8

ans = 8



$\frac{1+10}{2} = 5 \longrightarrow 1$

$\frac{6+10}{2} = 8 \longrightarrow 2$

$\frac{6+7}{2} = 6 \longrightarrow 3$

$\frac{7+7}{2} = 7 \longrightarrow 4$

```
while (L<=H)
    mid = (L+H)/2
    if (isBadVersion(mid) == true){
        ans = mid
        H = mid-1
    }else{
        L = mid+1
    }
return ans
```

2^5

2×2^4

$2^{31} - 7 + 2^{31} - 1$

L H

$\longrightarrow \frac{2^{31}-1}{2^{31}-4}$

$1 = 4B$

$-3/2$

L + H

$2^{31} - 1$

$\frac{2^{31} + 2^{31}}{2^{32}}$

$Lo + Hi$

$\boxed{mid = Lo + (Hi - Lo)/2}$

sort colors

nums = [0, 0, 1, 1, 2, 0]

o/p = [0, 0, 1, 1, 2, 2]

red white blue

0	1	2

i j k

[2, 0, 2, 1, 1, 0]

i j k

[0, 0, 1, 1, 2, 2]

i = 0 = [0] = 2
i = 0 = [0] = 0
i = 1 = [1] = 0
= 2 = [2] = 2
= 2 = [2] = 1
= 3 = [3] = 1
= 4 = [4] = 2

i = 0
j = 0
k = n-1

```
while(i <= k){
    if(nun[i] == 0){
        temp = nums[i]
        nums[i] = nums[j]
        nums[j] = temp
        j++
        i++
    } else if(nums[i] == 2){
        temp = nums[i]
        nums[i] = nums[k]
        nums[k] = temp
        k--
    } else {
```

i ++

}

}